Track BC v0.1 — Hydrogen 1s Shilov boundary condition: substrate-mechanism framework

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Track BC v0.1 — Hydrogen 1s Shilov boundary condition

1. The framework — Bergman integral formula

Per SPLP candidate (Memorial Day Half-Integer Axis G v0.1) + A_sub v0.4: physical setups specify boundary conditions on Shilov boundary $S^4 \times S^1 \subset \partial$D_IV^5. The substrate state is then reconstructed via Bergman reproducing kernel:

ψ(z) = $\int\{S^4 \times S^1\}\, K(z, \bar{w}) \cdot \varphi$\{boundary\}($\bar{w}$) d$\mu$($\bar{w}$)

where: - K(z, $\bar{w}$) = c_FK · (1 - z · $\bar{w}$)^(-7/2) is the Bergman kernel (T2442 RIGOROUSLY CLOSED) - φ_{boundary}($\bar{w}$) is the boundary-condition specification on Shilov boundary - dμ($\bar{w}$) is the Shilov boundary Hua-Look measure

For hydrogen 1s electron: identify φ_{boundary} explicitly + match resulting ψ to observed 1s wavefunction.

2. K-type identification — substrate electron and substrate proton

2.1 Substrate electron K-type

Electron quantum numbers: charge Q = -1, spin S = 1/2, chirality $\gamma^5 = \pm 1$ (both eigenvalues present in Dirac equation).

In substrate Wallach K-type basis: electron corresponds to half-integer Pin(2) cover K-type with Q̂ = -1. The lowest-Casimir such K-type (consistent with Pin(2)-positive-chirality projection) is:

V_(electron) = V_(1/2, -1/2) (Pin(2) cover, m_2 = -1/2 ↔ Q = -1 in suitable units)

Casimir eigenvalue: C_2(electron) = (1/2)(1/2 + n_C - 1) + (-1/2)(-1/2 + N_c - 2) = (1/2)(9/2) + (-1/2)(1/2) = 9/4 - 1/4 = 8/4 = 2

So C_2(electron) = 2 in substrate Casimir-2 units. (v0.1 honest scope: simplest identification; multi-week K-type identification work to refine — Toy 3531 Track P closure.)

2.2 Substrate proton K-type

Proton: charge Q = +1, structure as 3-quark bound state, spin 1/2, chirality content TBD per substrate identification.

In substrate Wallach K-type basis: proton corresponds to half-integer Pin(2) cover K-type with $\hat{Q} = +1$ + composite-3-quark structure. The simplest candidate K-type for the substrate proton:

V_(proton) = V_(3/2, +1/2) (Pin(2) cover, m_2 = +1/2 ↔ Q = +1; m_1 = 3/2 from 3-quark composite)

Casimir eigenvalue: C_2(proton) = (3/2)(3/2 + 4) + (1/2)(1/2 + 1) = (3/2)(11/2) + (1/2)(3/2) = 33/4 + 3/4 = 36/4 = 9

So C_2(proton) = 9 in substrate units. (v0.1 honest scope: simplest identification; T187 m_p/m_e = $6\pi^5$ provides empirical anchor for proton K-type at m_p ≈ 1836 m_e. C_2(proton)/C_2(electron) = 9/2 = 4.5; not directly 1836, but in BST the Casimir spectrum gives substrate-natural mass squared and 1836 emerges from substrate-tick discreteness + α^{BST primary} mechanism per T2476. Multi-week explicit derivation.)

2.3 Bound electron-proton substrate state

For a hydrogen atom (bound electron + proton), the substrate state is a 2-K-type bound combination:

V_(hydrogen, bound) = V_(electron) ⊗*substrate V*(proton)

where ⊗_substrate denotes substrate-tensor-product respecting U(1)_em charge sum = 0 (Q = -1 + Q = +1 = 0 charge-singlet) + total angular momentum coupling + spin coupling.

The substrate identifies the bound hydrogen substrate state as a specific K-type in the tensor product Hilbert space H^2(D_IV^5) ⊗ H^2(D_IV^5).

3. Substrate-mechanism for boundary condition imposition

3.1 The proton's Coulomb field as Shilov boundary condition

The proton's static electric charge Q_p = +1 creates a U(1)*em gauge field A_μ(r)* ∝ *1/|r|. In substrate language: the proton's K-type V*(proton) at position R_p in physical space imposes a U(1)_em phase rotation on substrate fields at distance r from R_p:

Phase rotation factor: exp(i · q · α · (function of r/r_proton))

where q is the test K-type charge, α is the fine-structure constant (= 1/N_max per T2447 RIGOROUSLY CLOSED), and r_proton is the proton's "effective Shilov boundary radius" (substrate-natural length scale).

This phase rotation IS the Shilov boundary condition imposed by the proton on the electron's substrate state.

3.2 Explicit boundary condition function

For the electron at distance r from proton, the boundary condition φ_{boundary}(w̄) on Shilov boundary $S^4 \times S^1$ is:

φ_{boundary}(w̄; r) = exp(i · Q̂*electron · α · f(r))* · ⟨*V*(electron) | Bergman boundary value⟩

where: - Q̂*electron = -1 (electron charge eigenvalue) - α = 1/N_max = 1/137 (fine-structure constant) - f(r) is the substrate-Coulomb potential profile (substrate-derived ~ 1/r at long range, modified at short range by substrate-tick discreteness) - Bergman boundary value is the standard limit of the K-type V*(electron) approaching the Shilov boundary

3.3 Reconstructing the 1s wavefunction

Apply the Bergman integral (Section 1) with the boundary condition (Section 3.2):

ψ_{1s}(z) = *∫{$S^4 \times S^1$} K(z, w̄) · exp(i · (-1) · α · f(r(w̄))) · V*(electron)(w̄) dμ(w̄)

This is the explicit Shilov boundary condition formula for hydrogen 1s electron. v0.1 framework level; explicit evaluation of the integral requires: 1. Substrate-Coulomb potential f(r) derived from substrate K-type field-theory (multi-week) 2. Bergman boundary value of V_(electron) on Shilov boundary (multi-week) 3. Explicit Bergman integral evaluation matching ψ_{1s}(r) = (1/$\sqrt{(\pi a\_0^3)}$) e^(-r/a_0) (multi-week)

4. Casimir spectrum matching with E_1s

4.1 Hydrogen 1s ground state energy

E_1s = -13.6 eV = $-\alpha^2$ · m_e $c^2/2$ = $-(1/137)^2$ · 0.511 MeV / 2 = -13.605 eV (Rydberg formula).

In BST units: E_1s/m_e c^2 = $-\alpha^2/2$ = $-1/(2 \cdot 137^2)$ = -1/37538.

4.2 Substrate-mechanism for 1s energy

Per SPLP candidate: E_1s should emerge as Casimir eigenvalue (or eigenvalue difference) of the substrate-tensor-product K-type V_(electron) ⊗*substrate V*(proton) under hydrogen boundary condition.

Substrate-mechanism candidate: - Unbound electron has Casimir C_2(electron) = 2 (substrate units) - Bound 1s electron under proton's Shilov BC has Casimir reduced by α^2-binding factor - E_1s ~ α^2 · m_e c^2 emerges from substrate-mechanism via [B̂, T̂_tick] vacuum-kicker channel (today step 9 finding) restricted to the hydrogen-bound K-type subspace

The α^2 factor in E_1s reflects the double-substrate-mechanism: one factor of α from substrate-tick scale (T2447 N_max = 1/α; T2476 α^{BST primary} pattern), one factor of α from Bergman 7/2 weighting on Shilov boundary integral. Two α factors compound → α^2.

(Multi-week explicit derivation. v0.1 establishes structural framework.)

4.3 Higher principal quantum number

For excited states $E_n = -\alpha^2/2 \cdot m_e c^2 / n^2$: - n = 1: 1s; lowest-Casimir bound K-type - n = 2: 2s, 2p; next-level K-types via Track A_sub commutator-edges - Higher n: Rydberg series

The K-type graph reaction-table (A_sub v0.4 Section 4) enumerates these transitions explicitly once boundary-condition imposition is fully derived.

5. The substrate-mechanism for BC imposition (general principle)

5.1 Generalization beyond hydrogen

The principle: **physical objects in 3D space (proton, nucleus, lattice, photon) IMPOSE Shilov boundary conditions on substrate K-types via their substrate K-type identification + U(1)_em + gauge field structure**.

For arbitrary physical setup: - Each physical object has substrate K-type identification (its Wallach K-type) - Each object's gauge charge content creates phase rotation on substrate boundary - The composite physical setup creates composite Shilov boundary condition - Substrate state = Bergman integral over composite boundary condition

This is the operational form of SPLP: each physical setup translates to an explicit Shilov boundary condition; observables are eigenvalues of substrate operators under this BC.

5.2 Connection to T2476 α^{BST primary} substrate-mechanism

Per T2476 STRUCTURALLY VERIFIED: α has substrate-mechanism via N_max + Dirac $Z\cdot\alpha=1$ critical limit. In Track BC language: α IS the substrate-tick scale at which boundary conditions are imposed. Each boundary condition's "strength" scales with α; the Coulomb potential's substrate-mechanism is α-quantized at the substrate-tick level.

5.3 Forward vs back-fit (Calibration #27 STANDING)

This framework derives hydrogen 1s wavefunction FROM substrate principles: - K-type identification (electron + proton from BST primaries) - Bergman kernel (T2442 RIGOROUSLY CLOSED) - Substrate-Coulomb potential (substrate-mechanism via T2476 α-quantization) - Boundary integral (standard Bergman reproducing kernel theorem)

NOT back-fit from observed ψ_{1s}. If multi-week derivation produces wrong ψ_{1s}, that's honest negative result. The framework is falsifiable by explicit Bergman integral computation.

6. Honest scope (Cal #27 STANDING)

What's RATIFIED / STRUCTURALLY VERIFIED in v0.1: - Bergman reproducing kernel K(z, w̄) ∝ (1 - z·w̄)^(-7/2) (RATIFIED standard math + T2442) - Electron / proton K-type identifications use substrate-natural BST primary content (v0.1 SIMPLEST identification; multi-week refinement) - α = 1/N_max (T2447 RIGOROUSLY CLOSED + Cal #126 + Cal #21) - Standard Bergman boundary-value theory (RATIFIED standard math)

What's FRAMEWORK in v0.1: - Bergman integral formula structure (Section 1) - K-type identification methodology (Section 2) - Substrate-Coulomb potential as substrate-mechanism for BC imposition (Section 3) - α^2-binding factor structural reading (Section 4.2) - SPLP operational form generalization (Section 5)

What's INTERPRETIVE in v0.1: - Section 2.1 V_(electron) = V_(1/2, -1/2) — simplest identification; multi-week Toy 3531 verification - Section 2.2 V_(proton) = V_(3/2, +1/2) — simplest identification; m_p/m_e = $6\pi^5$ matching requires explicit derivation - Section 3.2 substrate-Coulomb f(r) — substrate-derived ~ 1/r at long range; explicit form requires field-theoretic derivation - Section 4.2 double-α substrate-mechanism — structural reading; explicit derivation multi-week

What's NOT in v0.1 (multi-week work): - Explicit Bergman integral evaluation matching ψ_{1s}(r) - Substrate-Coulomb potential f(r) explicit derivation from substrate field theory - Substrate-tensor-product ⊗_substrate explicit structure for bound K-types - Substrate-mechanism for excited states (2s, 2p, n ≥ 2) - Substrate-mechanism for other bound states (helium, lithium, hydrogen molecule)

Cal #27 STANDING reflexive trigger count: 2 triggers (Section 2 K-type identifications + Section 4.2 double-α structural reading). Both flagged INTERPRETIVE in honest scope; framework-level rather than derived.

7. Connection to today's morning Lyra-lane work

This v0.1 absorbs: - A_sub v0.4 K-type graph reaction table (Section 1.1 framing): boundary conditions select K-types via reaction-table edges - 9 A_sub commutator closures (steps 1-9): algebraic structure underlying boundary-condition imposition - [B̂, T̂_tick] = β · |V_(1,0)⟩⟨V_(0,0)| vacuum kicker (step 9): mechanism for energy-level transitions in bound states - Half-Integer Axis G v0.2 partition (S^1-derived dynamics): boundary conditions impose phase content on S^1 Pin(2) cover - **{Q̂, P̂_op} = 0** (step 1): substrate-parity's charge-flip structure may affect bound-state K-type identification

The cumulative morning work establishes the algebraic foundation for Track BC's substrate-mechanism. v0.2+ multi-week work proceeds via Bergman integral evaluation + substrate-Coulomb derivation.

8. Recommended next pulls

1. Reaction table enumeration for lowest 20 K-types (A_sub v0.5 mechanical work): now tractable given 9 commutators closed. Feeds Track BC v0.2 (refines K-type identification) + Track P closure (Memorial Day Gap 1).
2. m_μ/m_e + m_τ/m_e dual-formula reconcile with Vol 1 Ch 11 + Vol 2 Ch 3: housekeeping joint with Elie; cross-volume cleanup per Cal #93+#94+#95.
3. Track DC v0.1 — explicit Bell 1/8 derivation from step 9 vacuum-kicker + step 1 simultaneous-diagonalizability violation: 8-sided die candidate mechanism explicit derivation. Multi-week substantial.
4. Vol 0 reader-grade polish (board item): multi-day; lower urgency.

9. Coordination

Elie: Toy 3531 (K-type population for leptons e/μ/τ) + Toy 3532 (Bergman integral for hydrogen 1s) joint with Lyra theoretical. Toy 3532 specifically tests Section 4.2-4.3 explicit calculation.

Grace: catalog entries for V_(electron) + V_(proton) K-type identifications + hydrogen 1s Shilov BC formula; cross-references to Track P (Memorial Day Gap 1) + Track DC (Gap 3) + SPLP candidate.

Cal: cold-read of Section 2 K-type identification methodology + Section 5 SPLP operational form generalization. Type C level-crossing per Cal #122.

Keeper: K-audit chain entry for Track BC v0.1 framework; promotion path via Cal cold-read + multi-week Bergman integral evaluation work.

— Lyra, Track BC v0.1 hydrogen 1s Shilov boundary condition framework filed Tuesday 2026-05-26 ~09:30 EDT per Memorial Day Gap 2 + Operational Physics Queue Track BC. FRAMEWORK grade; multi-week explicit derivation path identified. Absorbs morning A_sub v0.4 + 9-commutator-closure algebraic foundation. SPLP operational form generalized: physical setups translate to Shilov BCs via substrate K-type identification + α-quantized phase rotations.